Adrian Domínguez Castro, Ph.D.

Data Scientist & AI/ML Engineer | LLM Systems, RAG & Agentic Workflows | Quantitative Research

Email · Linkedin · GitHub · Google Scholar Website

Work Authorization: Permanent Resident (Green Card) / No sponsorship required

SUMMARY:

Quantitative Data Scientist & AI/ML Engineer (Ph.D. in Theoretical Physics) with 5+ years of experience designing scalable data pipelines, predictive models, and production-grade ML/GenAI systems. Combines a rigorous quantitative background (time series, regression, tree-based models) with modern LLM architectures, RAG and agentic workflows. Proven track record partnering with executive stakeholders to translate complex business challenges into high-impact data assets and technical solutions.

SELECTED EXPERIENCE

Principal AI/ML Engineering Contractor · Sept 2024-Present
Independent AI/ML Consulting

Partnered directly with executive stakeholders and cross-functional teams to translate complex business problem statements into technical AI/ML roadmaps and production-grade architectures.

Selected engagements:
Healthcare & research AI: Decimal.health | OncoBrain, AccelerOnc Studio, Moffitt Cancer Center

- Architected end-to-end hybrid RAG systems (Dense + BM25, Cohere Rerank) for unstructured oncology data, achieving high context retrieval precision across complex clinical records.
- Engineered automated ingestion pipelines using deep-parsing web scrapers and self-healing multi-agent validation loops, reducing data processing latency by 90%.
- Built autonomous AI agents using LangGraph with deterministic cyclic evaluation and self-correction loops for automated extraction, reducing structured extraction error rates by 30%.

Fintech AI: Stealth Mode Startup

- Engineered production-grade financial time series pipelines (LSTM Autoencoders, RNNs) for market price anomaly detection and forecasting, covering automated ingestion, feature engineering, and model validation; created open-source [WTI Oil Price Anomaly Detection] reference implementation.
- Evaluated modeling trade-offs (traditional ML vs. Deep Learning vs. GenAI) and built clean, production-grade Python code integrated into client infrastructure.

Postdoctoral Research Scientist · Feb 2023-Jan 2024
Vanderbilt University, USA

- Built a physics-informed ML pipeline (Random Forest, RNNs) on DFT data, achieving 85% predictive accuracy for material properties. [Link]
- Implemented automated Python data pipelines and predictive models (RNNs for time series, XGBoost for feature extraction) from simulation datasets on high-performance computing (HPC) infrastructure, reducing data processing latency by 80%.

Doctoral Research Scientist · Oct 2018-Dec 2022
University of Bremen, Germany

- Led 3+ multi-year research projects, engineered end-to-end data science workflows for complex scientific datasets, reducing processing time by 50% and generating 4 peer-reviewed publications.
- Built and led cross-functional research teams across five countries, directing large-scale predictive modeling and simulation workflows for complex molecular systems.

Associate Research Scientist · Sept 2017-Jul 2018
University of Groningen, The Netherlands

- Developed computational research and mentored junior scientists; built Python automation pipelines for large-scale data analysis (50% processing time reduction) and optimized high-performance simulation code.

EDUCATION

Ph.D. in Theoretical Physics · 2022 · University of Bremen, Germany

B.Sc. in Radiochemistry · 2014 · InSTEC-University of Havana, Cuba

AI SYSTEMS & TECHNICAL EXPERTISE

LLM & GenAI Systems: Fine-tuning, Transformers, Prompt Engineering, RAG, LangChain, LangGraph, Agentic Workflows, Vector Databases (ChromaDB, Qdrant), Model Evaluation, NLP.
Machine Learning & Modeling: Regression (LASSO, Elastic Net), Tree-Based Models (XGBoost, Random Forest), Time Series Analysis & Forecasting, Predictive Modeling, Statistical Analysis.
Deep Learning: Neural Networks (MLP, RNN, LSTM), Physics-Informed ML, Model Evaluation.
Programming & Infrastructure: Python (Production Code, PyTorch, Hugging Face, NumPy, Pandas, scikit-learn), C/C++, Shell, Linux, HPC, Automated Data Pipelines, Azure ML, GCP
Soft Skills: Executive & Senior Stakeholder Engagement, Cross-Functional Team Leadership, Business-to-Technical Translation, Agile Project Management

SELECTED PUBLICATIONS

- Featured Papers: Author/Co-author of 9 peer-reviewed publications in top-tier scientific journals (including Physical Review Letters, ACS Catalysis, Nanoscale, Theoretical Chemistry Accounts, and Communications Chemistry) focusing on quantum dynamics, density functional theory (DFT), and machine learning applications.
- Full List: Complete publication list and citation metrics available on Google Scholar Profile.